

A Selective Electromagnetic-Transfer Candidate for the Cosmological Lithium Problem
September 2026 GALAH posterior, direct LINX and external D-chain covariance update
Planck PR4/NPIPE, CMB blackbody and cross-network qualification retained

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EVIDENCE STATUS: Formal paper draft. The Planck PR4/NPIPE CMB posterior, CMB blackbody proxy, GALAH posterior propagation and 60-row direct LINX abundance gates are technically qualified.

SCIENTIFIC STATUS: Gate incomplete. Zero of 60 abundance-passing candidates lies inside the available partial Pisanti/LUNA 95% D-chain covariance region under the registered rate mapping.

PROMOTION BOUNDARY: ordinary quoted D-chain rate uncertainty is strongly disfavoured as the required explanation; a separately action-derived source remains unexcluded but needs a frozen action, exact LGLL or TUNL posterior products, and independent-network mediator propagation.

Description

The early Universe made mostly hydrogen and helium, plus tiny amounts of deuterium, helium-3 and lithium. Modern calculations predict deuterium and helium very well, but they predict too much lithium compared with old metal-poor stars.

This long-standing mismatch is termed the cosmological lithium problem.

The July tests tried broad ways to change the early-Universe calculation.

Those did not work: when lithium moved in the right direction, something else, usually deuterium or helium, moved out of agreement.

The August work then tested a more selective idea: a late, narrow electromagnetic process could reduce the mass-7 reservoir after normal nucleosynthesis without disturbing the successful deuterium and helium predictions.

The September update asks whether this selected candidate is still acceptable when compared with the detailed Planck CMB data, including calibration and foreground nuisance parameters.

The current answer is yes for the CMB gate, but the later nuclear-rate checks sharpen the boundary: the candidate remains conditional on D/H, lithium, helium and helium-3 behaviour under realistic nuclear-rate covariance and direct mediator-aware abundance propagation.

Latest Result in Plain Language

The newer stellar analysis makes the lithium target narrower. A model trained on GALAH DR3 and checked unchanged on GALAH DR4 places the least-depleted stellar-surface plateau at $A(\text{Li}) = 2.09062$, corresponding to $\text{Li}/\text{H} = 1.232 \times 10^{-10}$, with a 95% interval of $1.073\text{--}1.424 \times 10^{-10}$. This is not a direct measurement of primordial lithium, but it is a more explicit observational target than the earlier fixed band.

A large computer search then found many mathematical combinations that fit deuterium, helium, lithium and helium-3, and 60 selected combinations survived direct LINX propagation. The decisive new check asks whether the required deuterium-reaction changes are small enough to be explained by published nuclear-rate uncertainty. They are not: none of the 60 candidates enters the available 95% covariance region.

The current result therefore separates two explanations. Ordinary measurement uncertainty in the three D-sensitive nuclear reactions is strongly disfavoured as the source of the needed shift. A genuinely new mediator or source interaction is not ruled out by this covariance test, but it must now be derived from a declared physical action and reproduced in an independent BBN network before it can be promoted.

Connection to Existing Papers

This revision follows the stronger Physics Reports style in which external papers are integrated into the structure of the argument. The references below are not simply a reading list.

Each one defines either an observational anchor, a numerical-method benchmark, a physical safety gate or a boundary on what can be claimed.

The most important model for this structure is the 2026 BBN sensitivity-atlas review by Burns, together with its public atlas repository.

That work separates BBN into observational inputs, theoretical machinery, sensitivity coefficients, uncertainty budgets and illustrative applications.

The present BBN-Li manuscript uses the same logic but applies it to a narrower claim: whether the selective electromagnetic-transfer candidate survives the registered abundance, CMB, spectral and nuclear-rate gates.

Table 1a. External references as analytical gates and comparators

External reference	Link	Role in the argument	Effect on the current claim
Burns 2026 Physics Reports / BBN sensitivity atlas	Open source	Publication-style comparator: separates observations, theory framework, sensitivity coefficients, uncertainty budgets and illustrative lithium application.	The BBN-Li paper should mirror this logic: each external reference defines a gate or comparison, not merely a bibliography entry.
arXiv:2603.22414	Open source	Open version of the Burns BBN sensitivity review and table structure.	Provides a direct model for linking reaction sensitivities and uncertainty budgets to the lithium problem.
BBN sensitivity atlas GitHub	Open source	Public numerical atlas with response functions, uncertainty budgets and PRIMAT/NACRE-II rate-compilation comparisons.	Used as the main external response benchmark for the September 12 atlas import and NACRE-II stress branch.
Fields 2011 lithium review	Open source	Problem-definition reference for the primordial lithium anomaly.	Frames why a successful mechanism must reduce mass-7 without breaking the otherwise successful BBN light-element pattern.
Cybert Fields Olive Yeh 2016	Open source	Precision standard-BBN reference after CMB-era baryon-density constraints.	Defines why broad expansion or baryon-density retuning is a high-risk route and why D/H and helium gates are non-negotiable.
Pitrou Coc Uzan Vangioni 2018 PRIMAT	Open source	Precision BBN and PRIMAT reference, especially for helium and weak-rate treatment.	Supports keeping PRIMAT cross-network comparison separate from a purely local LINX response result.
LINX code paper	Open source	Differentiable BBN network and joint CMB/BBN computational route.	Supports the local response-derivative/eigenmode-validation strategy, while still requiring direct mediator-aware propagation.
LINX arXiv preprint	Open source	Open preprint and implementation description for Light Isotope Nucleosynthesis with JAX.	Supports the technical decision to use LINX for fast response scans and direct finite-pull reruns.
ACROPOLIS electromagnetic-cascade framework	Open source	External cascade-physics precedent for electromagnetic decays and light-element photodisintegration.	Defines why a late electromagnetic-transfer candidate needs cascade and threshold gates, not just a lithium-yield fit.
Planck 2018 cosmological parameters	Open source	CMB baseline for baryon density, cosmological parameters and compressed-background consistency.	Motivates replacing compressed CMB priors with the native PR4/NPIPE nuisance posterior used in September.
COBE FIRAS CMB spectrum	Open source	Spectral-distortion observational reference.	Defines the need for spectral-safety gates for any post-BBN electromagnetic energy release.
Khatri and Sunyaev 2012 CMB spectrum formation	Open source	Thermalisation and blackbody-photosphere physics reference.	Frames why mu-distortion and thermalisation checks are part of the candidate safety case.
CosmoVerse White Paper 2025	DOI	Broad review of H0, BAO, CMB, BBN, systematics, varying constants, dark-sector extensions and parity-sensitive early-Universe physics.	Added as an external gate reference: lithium improvement must survive D/H, Yp, He3/D, CMB, spectral, nuclear-rate and cross-dataset checks before promotion.
Wang et al. 2024 GALAH 3D-NLTE lithium catalogue	Open source	Large stellar-lithium catalogue used to build the empirical censored plateau/depletion model.	Defines the revised empirical stellar-lithium target and its limits.
Launders, Giovanetti and Liu 2026 deuterium GP analysis	Open source	Highest-priority source for synchronised, correlated D-chain functional uncertainty.	Defines the current external nuclear-rate promotion gate.
Pisanti et al. 2021, Primordial Deuterium after LUNA	Open source	Published coefficient/error-budget basis used for the executable partial external covariance gate.	Defines the current external nuclear-rate promotion gate.
Mossa et al. 2020 LUNA deuterium-burning measurement	Open source	Experimental d(p,γ)3He anchor underlying the improved deuterium-rate programme.	Defines the current external nuclear-rate promotion gate.
Gomez Inesta, Iliadis and Coc 2017 Bayesian d+d rates	Open source	Bayesian treatment of the paired d(d,n)3He and d(d,p)t reaction rates.	Defines the current external nuclear-rate promotion gate.
Moscato et al. 2021 Bayesian d(p,γ)3He rate	Open source	Independent hierarchical Bayesian rate analysis for the third D-sensitive reaction.	Defines the current external nuclear-rate promotion gate.

This reference-led structure changes the burden of proof. A lithium-only improvement is not sufficient. The candidate must also remain compatible with D/H, Yp, He3/D, the CMB baryon-density posterior, spectral-distortion limits, cascade physics and nuclear-rate uncertainty. Where an external reference is weaker, incomplete or compilation-dependent, the manuscript keeps that point as an open gate rather than promoting the model.

The CosmoVerse 2025 white paper is now included as a broad external gate reference. It does not treat lithium, H0, BAO, CMB, growth, varying constants or calibration as isolated problems.

For this BBN-Li candidate the August lithium cell must therefore be judged against D/H, Yp, He3/D, Planck/ACT/SPT CMB consistency, spectral safety, BAO+BBN calibration and local-systematics controls.

The September 16 update adds an explicit CMB blackbody validation gate.

This gate asks whether the same electromagnetic-transfer idea that may reduce mass-7 can remain compatible with the near-perfect FIRAS CMB spectrum, photon-number accounting, entropy conservation and CMB temperature-redshift tests.

The work sits inside a long literature on the lithium anomaly, precision BBN, stellar depletion, electromagnetic cascades, CMB constraints and spectral distortions.

The references listed below are used as the current interpretive frame.

They define the observational problem, the standard BBN baseline, the cascade machinery and the CMB/spectral gates.

Table 1. Literature cross-reference

Topic	Representative references	Role in this paper
Problem definition	Fields 2011; Cyburt, Fields and Olive 2008; Spite and Spite 1982; Sbordone et al. 2010; Ryan 2023.	Defines the Li-7 excess relative to metal-poor stellar plateaux.
Precision BBN	Cyburt et al. 2016; Pitrou et al. 2018; Fields et al. 2020; LINX 2024.	Defines standard abundance calculations and CMB-compatible baryon density handling.
Observational anchors	Cooke et al.; helium determinations; Planck Collaboration 2020.	Defines D/H, Y _p and CMB posterior constraints.
Cascade and late-decay physics	Depta et al. ACROPOLIS 2021 and related electromagnetic-decay constraints.	Tests whether a late electromagnetic process damages other light elements.
CMB and spectral gates	Planck PR4/NPIPE/CamSpec; Fixsen FIRAS; Khatri and Sunyaev.	Tests CMB consistency and spectral-distortion safety.

Abstract

A staged investigation of the cosmological lithium problem is reported in a reference-integrated form.

July 2026 broad BBN and cosmological controls produced no joint D/H, helium and Li-7 solution.

August work identified a selective post-BBN electromagnetic-transfer candidate, and September Planck PR4/NPIPE and CMB-blackbody audits retained that candidate under their registered gates.

The present update replaces the fixed stellar-lithium target with a censored GALAH DR3 3D-NLTE model, frozen before a GALAH DR4 validation.

The inferred least-depleted stellar-surface plateau is $A(\text{Li}) = 2.09062$ with 95% interval 2.03076-2.15345; this is an empirical surface anchor, not a primordial-abundance determination.

An adaptive 200,000-row hypothesis search yielded 138,797 abundance-gate passes, after which 60 selected mediator plus D-chain rows were propagated directly in LINX and all 60 passed the active GALAH abundance gate.

However, none lies inside the available partial Pisanti/LUNA 95% D-chain covariance region.

The best candidate has abundance $\chi^2 = 1.9079$ but external $D_M^2 = 136.914$, compared with a three-dimensional 95% threshold of 7.815.

The analysis therefore disfavours ordinary quoted nuclear-rate uncertainty as the required explanation under the registered mapping, while leaving a separately action-derived new-physics source untested. Exact LGLL Gaussian-process or paired TUNL posterior products and independent-network mediator propagation remain the principal promotion gates.

1. Introduction

The cosmological lithium problem is unusual because standard BBN is not globally broken.

Precision-BBN references and CMB-era reviews show that D/H and helium largely support the standard thermal history, while lithium-7 remains high relative to metal-poor stellar plateau observations.

The external literature therefore imposes the central constraint on this paper: a viable candidate must reduce the mass-7 reservoir without using a broad background retuning that spoils deuterium, helium, helium-3 or the CMB baryon-density posterior.

This paper records the transition from negative July controls, through the positive August candidate and September CMB qualification, to the September 23-24 observational-anchor and external-covariance tests. The chronology matters: the later tests narrow the earlier positive result rather than replacing or hiding it.

2. Methods

The computational programme used LINX for BBN network and abundance validation, ACROPOLIS for electromagnetic cascade checks, Cobaya and CAMB for native Planck CMB likelihood sampling, and project-specific Python audit layers for qualification gates and reproducibility.

The external literature determines how these modules are interpreted: LINX and PRIMAT define independent network checks, ACROPOLIS defines cascade safety, Planck/FIRAS define CMB and spectral constraints, and the Burns sensitivity atlas supplies an external response-function comparison for reaction-rate uncertainty.

Table 2. Method blocks

Method block	Implementation	Decision role
Standard and modified BBN controls	July LINX/FR/SU2 scripts.	Negative control and failure-boundary definition.
Selective transfer/cascade checks	ACROPOLIS plus LINX candidate pipeline.	Candidate discovery and abundance safety screen.
Finite-temperature production	Momentum-resolved and resummed production calculations.	Tests whether the candidate abundance is physically plausible.
Native CMB nuisance likelihood	Cobaya/CAMB with Planck PR4/NPIPE CamSpec, low-I TT, low-I EE and PR4 lensing.	Replaces compressed CMB gate.
Qualification audit	Split-chain, autocorrelation ESS, nuisance stability and registered-cell density scripts.	Determines whether the CMB gate is technically qualified.

3. Results

3.1 July Negative Controls

The July programme reproduced the anomaly and tested whether broad modifications could resolve it.

The result was negative: no tested global modification simultaneously preserved deuterium, helium and lithium.

This result is scientifically useful because it sharply constrains what a successful mechanism can look like.

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Traceability item	Recorded value
Stage	July 16-18, 2026.
Result	No D/H + helium + Li-7 pass from standard, FR/clock-proxy, reaction-rate or SU2-style control scans.
Consequence	Future solution must be selective rather than a broad background retuning.

3.2 August Selective Candidate

The August candidate uses a narrow electromagnetic window after ordinary BBN.

Its central physical premise is that mass-7, mostly Be-7 before electron capture, can be reduced without crossing into broad deuterium destruction.

The candidate therefore targets the unusual topology of the mass-7 pathway rather than changing the whole thermal history.

Table 3. Registered August candidate

Quantity	Value
mediator_mass_MeV	4.44
photon_energy_MeV	2.22
lifetime_s	1.584893e+05
n_phi_over_n_gamma_at_10MeV	1.211528e-06
g_phi_gamma_GeV_inv	9.767274e-11
D_H_times_1e5	2.493161
Yp	0.246735
Li7_H_times_1e10	1.488554
He3_D	0.415242
eta10_CMB	6.096517
Neff_CMB	3.044601
mu_distortion	1.476434e-09
joint_pass_probability	0.650158
delta_log_evidence_vs_standard	105.206012
Planck_bestfit_eta10	6.105374
Planck_bestfit_nnu	2.84265
Planck_bestfit_YHe	0.282034
Planck_bestfit_chi2_CMB	10971.285

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Traceability item	Recorded value
Stage	August 20-21, 2026.
Positive result	Candidate passes the recorded abundance/cascade/FIRAS gates at the frozen cell.
Boundary	A complete action-level particle model is outstanding.

3.3 Native Planck CMB Nuisance Qualification

The native Planck PR4/NPIPE test is the strongest new constraint in this update.

It lets ω_b , N_{eff} and Y_p vary independently, and marginalises over foreground and calibration nuisance directions rather than using only a compressed two-variable Planck prior.

Table 4. Qualification gates

Gate	Threshold	Observed	Pass
Cobaya split-chain mean R-1	≤ 0.03	0.0293	True
Independent split R-1 max over ombh2, nnu, YHe, nuisance	≤ 0.03	0.0021	True
Autocorrelation ESS min over ombh2, nnu, YHe, nuisance	≥ 1000	1449.0004	True
Registered lithium cell joint posterior density	$\Delta\chi^2 \leq 7.814728$	1.2574	True
Nuisance split stability diagnostic	max split mean shift ≤ 1 sigma	0.0838	True
First-half vs second-half stability diagnostic	max mean shift ≤ 1 sigma	0.0926	True

Table 5. Registered-cell posterior density

Quantity	Value
posterior mean η_{10}	6.074645
posterior mean N_{eff}	2.814036
posterior mean Y_p	0.262997
registered η_{10}	6.096365
registered N_{eff}	3.044709
registered Y_p	0.246733
registered $\Delta\chi^2$	1.25743
Mahalanobis distance	1.121352
density ratio to posterior peak	0.533277
posterior mass inside equal-density contour	0.261374
joint 95 gate pass	True
weighted kNN density ratio	0.601247

Table 6. CMB posterior summary

Parameter	Mean	Std	Median	68% interval	95% interval
ombh2	0.02219	1.9507e-04	0.02219	0.021995 to 0.022381	0.021809 to 0.022573
nnu	2.81855	0.2532	2.81194	2.569554 to 3.062147	2.343549 to 3.327882
YHe	0.262427	0.0186	0.262668	0.244057 to 0.280933	0.225078 to 0.298364
η_{10}	6.075084	0.0534	6.075231	6.021877 to 6.127459	5.97098 to 6.179927
H0	66.210921	1.5058	66.164438	64.706901 to 67.68329	63.421067 to 69.23348
omegam	0.31734	0.0087	0.317346	0.308729 to 0.326008	0.300514 to 0.334223
sigma8	0.803256	0.0095	0.803151	0.793847 to 0.812604	0.784934 to 0.822145
S8	0.826006	0.0111	0.826026	0.814952 to 0.837015	0.804369 to 0.847849
χ^2_{CMB}	10984.919918	7.9961	10984.146	10979.26564 to 10990.31448	10975.553 to 10997.627775

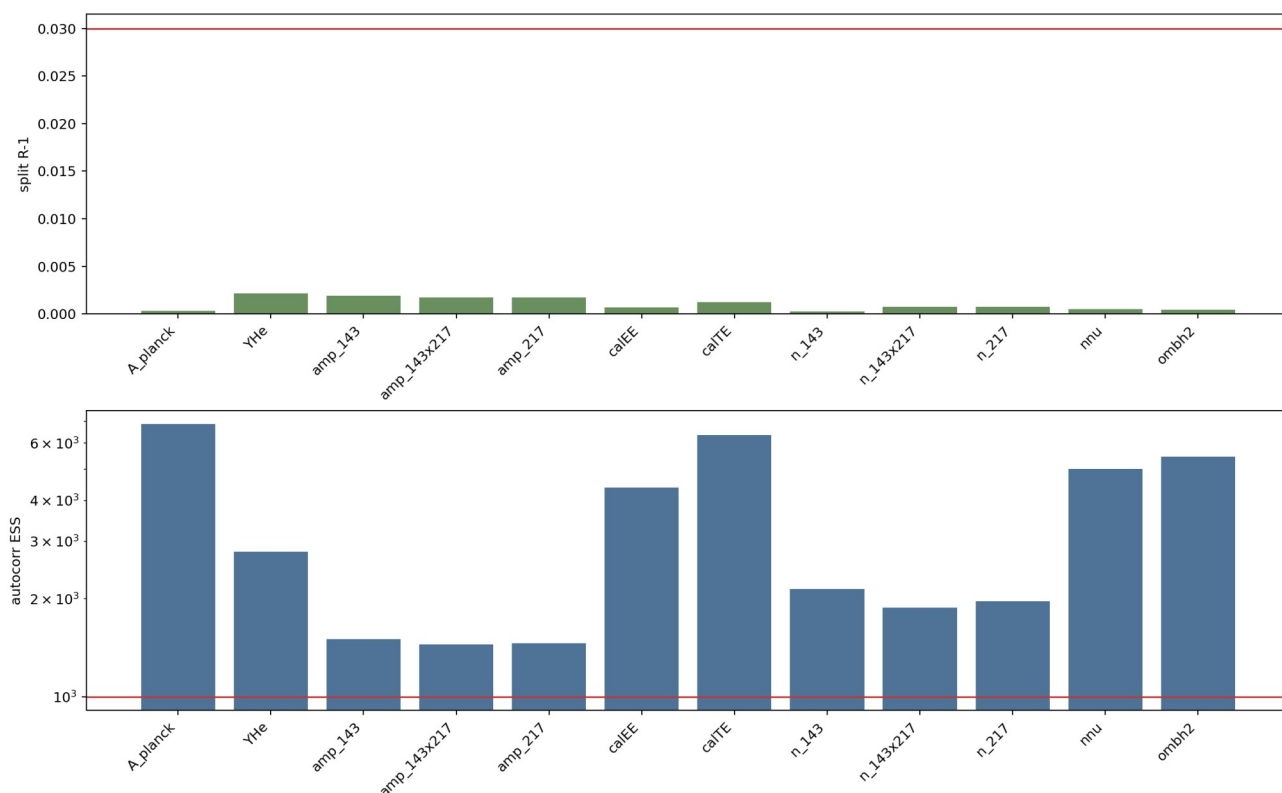


Figure 1. Combined main plus ESS top-up qualification gates.

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Traceability item	Recorded value
Stage	September 11, 2026.
Combined chain size	120073 compressed rows; raw weight sum 393272.
Gate outcome	PASS on all preregistered gates.
Most conservative ESS	1449 for amp_143x217 among audited gate parameters.

3.4 September 2026 NLO and Anchor-Sensitivity Updates

After the combined Planck PR4/NPIPE qualification audit, three September checks were added to test whether the August candidate survives extended sources of modelling uncertainty: a matched NLO thermal gate, a generic abundance-anchor sensitivity grid, and a named literature-anchor stress test.

These checks are not a new discovery claim; they are designed to state exactly which abundance assumptions leave the candidate viable.

- Matched NLO thermal gate:** 625/625 pass.
 Weighted NLO shift stayed within +/-2.569%; maximum single-cell total shift 2.741%; pole-cut consistency and thermal self-energy proxies remained inside the declared gate.
- Generic anchor sensitivity:** 486/1152 full pass; 15/18 registered-anchor pass.
 Failures were concentrated in the low-D/H plus high deuterium-destruction direction, showing that the candidate is sensitive to the deuterium anchor rather than freely adjustable.
- Named literature-anchor stress test:** 1020/2160 full pass; 15/18 registered literature/project cells pass.
 The registered cell remains viable, while high single-system D/H, EMPRESS-low helium and Izotov-high helium stress choices fail in this simplified gate.

The practical interpretation is narrower but stronger: the August candidate does not disappear under the September gate checks, but it should be written as conditional on the declared D/H, helium and nuclear-rate assumptions until the same point is reproduced inside an independent BBN solver with full nuclear covariance propagation.

3.5 September 12 Nuclear Rate and Cross Network Validation

After the September 11 paper draft, the nuclear-rate branch was expanded to test whether the registered lithium candidate is limited by software response, cross-network disagreement, observational anchors or the absence of a true reaction-by-reaction covariance source.

These checks are ordinary-BBN rate-pull diagnostics and response-propagation tests; they do not yet inject a physical mediator directly into PRIMAT, PArthENoPE or an independent network.

Table 7. September 12 nuclear-rate and direct-network validation summary

Check	Quantitative result	Interpretation
PRIMAT N=5000 rate Monte Carlo bridge	5000 samples; 12/12 mapped reactions; He3/D propagated; pass fraction 0.6036; median abundance χ^2 4.6314.	Covariance sampling is stable enough for a diagnostic, but the registered abundance gate remains incomplete.
N=1000 versus N=5000 covariance stability	Pass fraction moved from 0.6116 to 0.6036; Frobenius covariance drift 0.1158; dominant failures through D/H.	Increasing sample size did not rescue the gate. D/H is the limiting abundance channel.
Direct LINX eigenmode validation on N=5000 covariance basis	32/32 direct LINX runs completed; maximum D/Yp/Li response error 0.01023 observational-sigma units; He3/D max absolute error 0.0001677.	The local first-order response approximation remains technically validated along the tested covariance eigenmodes.
BBN sensitivity-atlas and NACRE-II stress branches	Five million draws per scenario; PRIMAT-source independent pass fractions near 0.608; NACRE-II independent pass fractions near 0.170.	PRIMAT-like responses align with the local LINX bridge; NACRE-II behaves as a strong rate-compilation stress branch.
Direct PRIMAT abundance reruns against reused exact LINX points	51/51 matched points completed; maximum direct D/Yp/Li disagreement 0.01137 observational-sigma units.	Cross-network disagreement is not the dominant failure mode at the tested ordinary-BBN rate-pull points.
Fresh exact LINX D-chain direct reruns	31/31 direct points completed; worst disagreement 0.02673 observational-sigma units at D_chain_all_positive_3, dominated by Li7/H; D/H grid pass fraction 0.54996.	Even the deliberately stressed D-chain direct points agree well between LINX and PRIMAT; the open gate is covariance and mediator physics, not direct-network mismatch.

The main inference is therefore narrower and cleaner than before.

The direct-network and response-linearity checks support the computational bridge, but the rate-covariance gate is still not closed because a large share of rate draws miss the registered joint abundance cell, especially through D/H. The most important failure-associated reaction directions remain dpHe3g, ddHe3n and ddtg.

This section should be read as a qualification update. It strengthens the evidence that the September BBN-Li programme is internally reproducible across LINX and PRIMAT at the tested points, while preserving the requirement for a published or package-native nuclear-rate covariance input and a direct physical mediator propagation test.

3.6 Reference Led Claim Gates

The external literature can be read as a set of claim gates. Problem-definition papers require a mass-7 reduction large enough to reach the observed lithium plateau.

Precision-BBN and CMB papers require D/H , Y_p , η_{10} and N_{eff} to stay inside their accepted ranges.

Cascade and spectral-distortion papers require late electromagnetic energy transfer to avoid deuterium destruction and blackbody-spectrum violations.

Sensitivity-atlas and rate-compilation papers require the result to survive reaction-rate uncertainty and PRIMAT/NACRE-II comparison.

Gate	External basis	Current BBN-Li status	Next required action
Lithium anomaly	GALAH DR3 3D-NLTE with Gaia controls; locked DR4 release validation; Fields/Spite context.	Target sharpened: $A(\text{Li}) = 2.09062$ [2.03076, 2.15345]; 3/46 earlier candidates survive the qualified posterior gate.	Replicate with an independent stellar survey and a physical stellar-depletion calculation.
D/H and helium preservation	Direct LINX candidate propagation plus the registered D/H, Y_p and He3/D anchors.	Technically satisfied for the selected 60-row queue, but the queue was deliberately enriched for abundance success.	Test the source in an independent network and under external joint D-chain posterior draws.
CMB baryon-density compatibility	Planck PR4/NPIPE nuisance posterior, compared against Planck-era cosmology.	Qualified under the combined main plus ESS top-up audit.	Extend to ACT/SPT where likelihood consistency allows.
Cascade and spectral safety	ACROPOLIS, FIRAS and Khatri/Sunyaev thermalisation physics.	Retained in recorded proxy gates; not yet evaluated for a fully action-derived source history.	Propagate one frozen source through matched NLO thermal, cascade and abundance calculations.
Nuclear-rate uncertainty	Pisanti/LUNA partial coefficient covariance; LGLL GP and TUNL Bayesian posterior targets.	Failed for ordinary uncertainty under the current mapping: 0/60 inside the available external 95% region.	Obtain exact synchronised LGLL draws or paired TUNL posterior samples; do not substitute the internal proxy.

This is the reason the present claim remains deliberately narrow.

The result is stronger than a lithium curve fit because it is cross-checked against external observational and computational structures.

It is still not a final solution because the external structures also identify the remaining failure modes.

3.7 CosmoVerse Tension Review Gate

The CosmoVerse review gives a useful check on the interpretation of the September BBN-Li programme.

It supports treating the lithium anomaly as real and persistent, but it also keeps stellar depletion, observational systematics and nuclear-rate uncertainty in play.

The present result therefore remains a candidate mechanism under test rather than a claimed resolution.

For the wider PLAMB and SU2R programme, the same review acts as a warning against single-dataset promotion.

A raw-MU or distance-law improvement must survive BAO geometry and CMB recombination constraints, while an SU2R background improvement must survive a perturbation and high-resolution CMB closure path.

Those requirements are now treated as linked gates rather than optional follow-up work.

Programme branch	CosmoVerse issue	Current interpretation	Next gate
BBN-Li	Lithium, D/H, Yp, He3/D and nuclear-rate uncertainty must be considered jointly.	The August cell remains promising, but September covariance and direct-network work keep the scientific gate incomplete.	Physical mediator-aware abundance propagation and a published or package-native covariance source.
PLAMB/FR	H0, BAO, CMB and local-distance-ladder systematics are cross-coupled.	Raw-MU linearity is not enough by itself; no-expansion claims must pass standard-ruler and recombination tests.	BAO+BBN consistency, distance-duality/opacity checks, redshift-frame and local-flow audits.
SU2/SU2R	Dark-sector extensions need growth, perturbation and CMB consistency, not only background-distance fits.	Effective-fluid status is retained; canonical non-phantom SU2R collapses toward LCDM in the current DESI-era run.	Freeze action-level route before direct ACT/SPT or perturbation-likelihood promotion.
Chiral diagnostics	Parity violation, primordial magnetic fields and chiral gravitational waves are relevant early-Universe probes.	These are useful diagnostics, but not direct late-time likelihood evidence for BBN-Li or SU2R.	Parity-observable likelihoods and CMB EB/TB or four-point-function hooks.

This gate improves the paper's publication structure because it ties the candidate mechanism to the wider observational-tension landscape.

It also preserves the negative result boundary: the BBN-Li candidate has not yet cleared the final covariance or mediator-aware propagation gates.

3.8 CMB Blackbody Validation Gate

The BBN-Li candidate must now pass an independent CMB spectral-shape gate.

A mechanism that selectively changes the late electromagnetic energy budget can reduce lithium in an abundance table but still fail cosmology if it distorts the CMB away from a Planck blackbody, changes photon number after light-element freeze-out, or shifts the baryon-to-photon ratio used by the CMB posterior.

The September 16 FR/CMB diagnostic is deliberately conservative.

It is not an official FIRAS covariance likelihood.

It tests representative Planck-preserving, grey, non-grey, dispersive and temperature-mixture transport classes on a 30-3000 GHz grid and refits the best grey amplitude and blackbody temperature before scoring residuals.

The result is a useful gate-setting scale: blackbody-preserving transport remains viable at the shape level, while non-grey and dispersive variants become constrained quickly.

Validation test	What it checks	Implication for BBN-Li
Thermalisation epoch classification	Classify whether mediator electromagnetic transfer occurs before the blackbody thermalisation surface, during the μ/γ distortion era, or after the CMB spectrum is effectively frozen.	A pre-thermalisation transfer is safer; a late transfer must predict sub-FIRAS non-grey, μ/γ and temperature-mixture residuals.
FIRAS-style spectral-shape projection	Project any photon-sector transfer into Planck-curve residuals after a best grey-amplitude and temperature refit.	The current 50 ppm RMS proxy gives a quick exclusion scale and must be replaced by an official FIRAS covariance likelihood before promotion.
Photon number and entropy accounting	Track whether the proposed lithium-sector transfer changes photon number, photon energy density, or η_B/γ after light-element freeze-out.	D/H, Y_p and the CMB baryon-density posterior are sensitive to entropy or photon dilution, so lithium-only tuning is insufficient.
Direct mediator-aware abundance propagation	Run the mediator model through LINX and an independent network rather than applying a response-level abundance shift.	This remains the main open scientific gate: current direct LINX/PRIMAT checks validate ordinary rate-pull plumbing, not the mediator dynamics.
CMB temperature-redshift consistency	Check whether any FR or photon-transport law preserves the Planck occupation structure and the observed $T(z)$ relation.	A model can preserve blackbody shape only if $h\nu / kT$ is transported coherently; non-grey or dispersive variants are tightly bounded.

The current 50 ppm RMS proxy limits, after best grey-amplitude and temperature refit, are:

Distortion family	Parameter	Approximate allowed amplitude	BBN-Li interpretation
Non-grey photon-transfer tilt	epsilon_tilt	3.27067597e-05	A frequency-dependent brightness or clock-rate tilt must be below this proxy amplitude after best blackbody refit.
Dispersive redshift remap	epsilon_dispersion	4.43003370e-07	A frequency-dependent redshift remapping is very tightly constrained because it warps the Planck curve.
Line-of-sight temperature mixture	sigma_lnT	4.02746473e-04	Path-dependent thermal or clock scatter cannot superpose a broad distribution of blackbodies without visible residuals.
Chemical-potential distortion	mu	1.43389573e-04	Standard mu-type spectral-distortion reference scale for energy release before full late thermalisation is lost.
Compton y distortion	y	8.48604116e-08	Standard y-type reference scale for late Comptonisation-like distortion in the same proxy gate.

Interpretation: this gate does not reject the August candidate cell, because the candidate has not yet been propagated as a fully specified photon-transfer history. It does, however, narrow the safe theory space.

A successful mechanism should either thermalise early enough to preserve the Planck occupation structure, or produce residual mu, y, non-grey and temperature-mixture amplitudes below FIRAS-level limits.

It should also leave D/H, Yp, He3/D and the CMB baryon-density posterior mutually consistent.

Linked program outputs: [diagnose_fr_cmb_blackbody_transform.py](#); [diagnose_fr_cmb_distortion_limits.py](#); summary outputs under plamb_runs/diagnostics/fr_cmb_blackbody_transform and plamb_runs/diagnostics/fr_cmb_distortion_limits.

3.9 GALAH Recalibration, Direct LINX Propagation and External D-Chain Covariance

The September 23-24 programme applies three distinct evidential layers. The GALAH analysis changes the observational lithium likelihood; the adaptive search and direct LINX batch test whether abundance-compatible candidate rows exist; the external covariance audit asks whether those rows are compatible with independently reported nuclear-rate uncertainty. These layers must not be collapsed into one pass fraction.

Table 8. September 23-24 evidence chronology

Stage	Design	Quantitative result	Interpretation boundary
23 Sep: GALAH DR3 model	977 stars; 734 detections and 243 upper limits; Gaia-informed censored mixture model.	$A(\text{Li}) = 2.09062$, 95% interval 2.03076-2.15345; max split R-hat 1.06886; min ESS 1334.7.	Technical pass; empirical stellar-surface anchor, not a primordial abundance determination.
23 Sep: locked GALAH DR4 check	No coefficient, prior, cut or imputation value refitted after opening DR4.	168 strict Gaia-source non-overlap stars; detection bias +0.0502 dex.	Useful release check, but not independent-survey replication.
23 Sep: archive re-score	277 CSVs audited; 70 compatible files; 244,867 complete, de-duplicated headline rows.	GALAH-posterior passes 21,737 versus 89,285 under the old fixed anchor; 24.35% retained.	Reclassification result; it does not rerun network dynamics.
23 Sep: adaptive shotgun	200,000 hypothesis-search rows using mediator and D-chain response structure.	138,797 posterior-gate passes; 56,431 pure-template passes; 5,529 within the tested internal q envelope.	Search enrichment only; pass fractions are not evidence probabilities.
24 Sep: direct LINX propagation	60 selected mediator plus D-chain rows propagated in LINX.	60/60 technical completions and 60/60 active GALAH abundance-gate passes; best $\chi^2 = 1.83743$.	Direct in LINX, but still an operator-split source prototype and not an independent-network mediator calculation.
24 Sep: external covariance audit	The same 60 rows were tested against the available partial Pisanti/LUNA three-reaction covariance.	0/60 inside the 95% region over $T_9 = 0.1\text{--}1.5$; best $D_M^2 = 136.914$ versus threshold 7.815.	Strong negative result for ordinary quoted D-chain uncertainty under the registered mapping.

The direct 60-row queue was selected from a search designed to find successful abundance combinations. Consequently, 60/60 direct abundance passes demonstrate reproducibility of those selected candidates in LINX; they do not estimate a prior predictive probability or a discovery significance.

Table 9. Best abundance-passing candidate under the external covariance audit

Quantity	Value
Candidate	combined_gaussian_T32_g2p25_s0p65__q_-1p5712_-2p3568_-2p3568
Direct abundance outputs	$D/H = 2.530419\text{e-}5$; $Y_p = 0.247566$; $\text{Li}/\text{H} = 1.393770\text{e-}10$; $\text{He3}/\text{D} = 0.404709$
Abundance χ^2	1.907920
Most favourable tested temperature	$T_9 = 0.4$
External Mahalanobis statistic	$D_M^2 = 136.913980$; $D_M = 11.701025$
External 95% threshold	$D_M^2 \leq 7.814728$ for three effective rate dimensions
Tail probability	1.74908e-29
Classification	ABUNDANCE_PASS_EXTERNAL_COVARIANCE_FAIL

The rate-pull coordinate q_i was converted to a physical fractional rate displacement using the PRIMAT temperature-dependent factor uncertainty $f_i(T_9)$:

$$\delta_{\text{Li}}(T_9) = \exp[q_i \ln f_i(T_9)] - 1 \quad (44)$$

Compatibility with the external fractional-rate covariance C was then scored by

$$D_M^2 = \delta^T C^+ \delta \quad (45)$$

where C^+ is the Moore-Penrose inverse.

The audit evaluated $T_9 = 0.1, 0.2, 0.4, 0.8, 1.0$ and 1.5 and reported the smallest distance, deliberately giving each candidate its most favourable tested mapping.

Table 10. Best candidate D-chain displacements at $T_9 = 0.4$

Reaction	Code label	Required shift	External 1-sigma	Marginal z
$d(p,\gamma)3\text{He}$	dpHe3g	-3.2126%	1.2146%	-2.645
$d(d,n)3\text{He}$	ddHe3n	-2.5454%	0.3714%	-6.853
$d(d,p)t$	ddtp	-2.5454%	0.2795%	-9.108

The d+d branches dominate the incompatibility. Their required shifts are only a few per cent in absolute terms, but the available external marginal uncertainties are below four-tenths of one per cent. Subsequently a modest rate displacement results in a large covariance distance.

Table 11. External D-chain posterior source hierarchy

Priority	Source	Current status	Required action
1	LGLL 2026 Gaussian-process posterior	Exact synchronised function/rate draws not yet released locally.	Request common-draw S-factor functions or thermally averaged rate draws on a shared T9 grid.
2	TUNL Bayesian D-rate posteriors	Published marginal analyses located; paired samples are not available locally.	Obtain paired d+d samples and a compatible d(p,gamma)3He chain.
3	Pisanti/LUNA coefficient covariance	Available and tested as independent per-reaction blocks; cross-reaction covariance is absent.	Retain as a partial external gate, not a complete joint posterior.
4	ETR25/RateMC archive	Archive checksum verified; no applicable D-chain files found.	Use for marginal-rate validation only if matching reaction products become available.
5	IAEA EXFOR reconstruction	Not yet built.	Preregister a joint experimental likelihood only if posterior products cannot be obtained.

Scientific interpretation: the present external test is partial because it combines independent per-reaction coefficient-covariance blocks and lacks released cross-reaction and shared-normalisation covariance.

Even with that favourable limitation, the best candidate is far outside the 95% region.

The result strongly rejects an ordinary-error explanation under this mapping.

It is not a no-go theorem for a new interaction that physically modifies effective reaction flow; that alternative must be judged from its action and source dynamics, not from the standard-rate covariance.

TV index

TV item	Assessment for the September 23-24 update
TV1-TV3: problem, observable, model	Observable gates are explicit; the source remains phenomenological rather than action-derived.
TV4-TV6: assumptions, mathematics, computation	Rate mapping, covariance statistic, temperature grid and direct LINX path are reproducible.
TV7-TV9: uncertainty, controls, falsification	The external covariance is a genuine falsifying control and returns a clear negative result for ordinary rate uncertainty.
TV10-TV12: comparison, novelty, promotion	Novelty is the linked observational-anchor/direct-propagation/external-covariance chain; promotion remains blocked.

4. Discussion

The latest audits change the scientific balance. The GALAH posterior and direct LINX propagation make the abundance target and numerical candidate family more explicit.

The external covariance audit then provides the strongest negative control so far: abundance success is not reproduced by rate shifts that look like ordinary published uncertainty.

This is useful evidence because it prevents a technically successful search from being mistaken for a nuclear-physics explanation.

The remaining physically meaningful route is narrower.

A new mediator or source may still change effective reaction flow, but that claim requires an action-level declaration of fields, interactions, conservation laws, thermal history and priors.

The same frozen source must then be propagated through LINX and at least one independent network, with CMB, FIRAS, D/H, Yp, He3/D and stellar-depletion alternatives retained as controls.

5. Limitations

- The analysis is not yet an independent external reproduction, and the new reference-integrated structure makes this limitation more explicit rather than hiding it.
- The mediator model is phenomenological; a fully specified action, symmetry content and particle-production history remain pending.
- The Planck run is spectrum-level but not map-level, and does not include ACT or SPT spectra.
- The abundance and CMB gates are joined by a registered cell rather than by a single global sampler over all microscopic nuisance directions.
- Systematic uncertainty in the stellar lithium plateau remains a separate astrophysical issue.

- The GALAH DR4 check is a locked release validation within the same survey family, not an independent stellar survey.
- The Pisanti/LUNA covariance is partial and block-diagonal; exact LGLL GP draws and paired TUNL posterior samples remain unavailable locally.
- The 60-row direct queue was selected for abundance performance, so its 100% pass rate cannot be read as a model probability.

6. Conclusions

The July investigations failed to resolve the lithium problem with broad BBN or cosmological modifications.

The August investigation found a selective electromagnetic-transfer candidate, and the Planck PR4/NPIPE and CMB-blackbody audits retained it under their declared gates.

The September GALAH analysis narrowed the observational lithium target, the adaptive search located candidate basins, and 60 selected combinations survived direct LINX abundance propagation.

The external covariance test is nevertheless decisive for the ordinary-rate interpretation: 0/60 candidates lies within the available partial Pisanti/LUNA 95% region, and the best has $D_M^2 = 136.914$.

The present paper therefore reports a constrained candidate and a scientifically important failed mechanism gate, not a solution of the cosmological lithium problem.

7. Future Work

1. Extend and independently reproduce the matched next-to-leading-order thermal corrections at and around the registered candidate cell.
2. Reproduce the ACROPOLIS/LINX candidate with independent code and fixed public manifests.
3. Extend CMB checks to ACT/SPT where likelihood access and parameter consistency allow.
4. Publish a frozen reproducibility bundle with checksums for scripts, input tables, plots and DOCX/PDF outputs.
5. Promote the model only after an action-level declaration specifies fields, interactions, gauge content, conservation laws and priors.
6. Replace the empirical PRIMAT rate-pull covariance proxy with a published, package-native or explicitly documented reaction covariance matrix.
7. Run a direct mediator-aware abundance propagation in LINX and at least one independent network, rather than applying the mediator cell as a response-level abundance shift.
8. Keep NACRE-II as a separate stress branch until reaction normalisations, response definitions and covariance assumptions are matched.

Highest-priority next tests

9. Obtain synchronised LGLL Gaussian-process S-factor or thermally averaged rate draws for all three D-sensitive reactions on a common T9 grid.
10. Obtain paired TUNL posterior samples and preserve shared experimental normalisation draws across the d+d branches.
11. Freeze an action-level mediator/source declaration before further source tuning, including fields, couplings, detailed balance, energy accounting and priors.
12. Propagate that frozen source through LINX and an independent network such as PRIMAT or PArthENoPE without response-level abundance shifts.
13. Replicate the GALAH lithium posterior with an independent stellar survey and a physical stellar-evolution depletion model.
14. Retain the existing Planck, FIRAS, D/H, Yp and He3/D gates unchanged during the mechanism test.

Appendix A. Acronyms

Table A1. Acronym glossary

Acronym	Meaning
ACROPOLIS	Public electromagnetic-cascade framework used for BBN photodisintegration constraints.
AIC	Akaike Information Criterion; lower is preferred after a penalty of 2 parameters.
BBN	Big Bang Nucleosynthesis.
BIC	Bayesian Information Criterion; lower is preferred with a stronger log(N) parameter penalty.
CAMB	Code for Anisotropies in the Microwave Background.
CMB	Cosmic microwave background.
Cobaya	Cosmological Bayesian analysis package used for the Planck nuisance MCMC.
D/H	Deuterium-to-hydrogen abundance ratio.
ESS	Effective sample size; here the autocorrelation-aware number of effectively independent posterior samples.
FIRAS	Far Infrared Absolute Spectrophotometer on COBE.
LINX	Fast differentiable BBN package used for abundance calculations and validation.
MCMC	Markov-chain Monte Carlo.
NPIPE	Planck reprocessing pipeline used for PR4 products.
PR4	Planck public release 4.
SPD	Symmetric positive definite; used for covariance handling and Cholesky/LAPACK inversion.
TV index	Traceability/validation index. The reports place it after each investigation section.
NLO	Next-to-leading order; the first correction beyond the leading thermal calculation.
EMPRESS	Extremely Metal-Poor Representatives Explored by the Subaru Survey; source of a low-helium stress anchor.
PRIMAT	Public precision BBN code used as an external comparison target for future reproduction.
PArthENoPE	Public BBN code used as an external comparison target for future reproduction.
GALAH	Galactic Archaeology with HERMES stellar spectroscopic survey.
GP	Gaussian process; a distribution over functions used here for correlated nuclear S-factor inference.
LGLL	Launders, Giovanetti and Liu 2026 deuterium Gaussian-process analysis.
LUNA	Laboratory for Underground Nuclear Astrophysics at Gran Sasso.
TUNL	Triangle Universities Nuclear Laboratory.

Appendix B. Variables

Table B1. Variable glossary

Variable	Meaning
eta10	10^{10} times the baryon-to-photon ratio η ; converted from ω_b using $\eta_{10} = 273.78 \omega_b$ for $T_{\text{CMB}} = 2.7255 \text{ K}$.
ω_b / ombh^2	Physical baryon density $\Omega_b h^2$.
ω_c / omch^2	Physical cold dark matter density $\Omega_c h^2$.
$N_{\text{eff}} / n_{\text{nu}}$	Effective relativistic species count inferred independently in the CMB nuisance chain.
Y_p / Y_{He}	Primordial helium-4 mass fraction; sampled independently in the full CMB likelihood.
D/H	Deuterium abundance anchor; required to remain within the adopted observational gate.
Li-7/H	Lithium-7 abundance. Most primordial mass-7 is produced as Be-7 and later decays to Li-7.
m_{phi}	Mediator mass in MeV for the selective electromagnetic-transfer candidate.
E_{gamma}	Characteristic photon energy, approximated as $m_{\text{phi}}/2$ for two-photon decay.
τ_{phi}	Mediator lifetime in seconds.
$n_{\text{phi}}/n_{\text{gamma}}$	Mediator abundance relative to photons at $T = 10 \text{ MeV}$.
$g_{\text{phi gamma}}$	Effective mediator-photon coupling in GeV^{-1} .
μ	CMB spectral-distortion chemical-potential style readout.
R-1	Convergence diagnostic used by Cobaya and the split-chain audit.
chi2	Goodness-of-fit statistic; lower values indicate a better fit for the same data and model complexity.
A(Li)	Logarithmic stellar lithium abundance, $A(\text{Li}) = \log_{10}(\text{Li}/\text{H}) + 12$.
Y_p	Primordial helium-4 mass fraction used by named helium abundance anchors.
τ_{int}	Integrated autocorrelation time used in the strict posterior qualification audit.
ESS	Effective sample size; the approximate number of independent posterior draws after autocorrelation.
q_i	Dimensionless pull applied to reaction i in the registered D-chain response parameterisation.
$f_i(T_9)$	Temperature-dependent multiplicative one-sigma factor uncertainty for reaction i .
$\delta_i(T_9)$	Physical fractional rate shift $\exp[q_i \ln f_i(T_9)] - 1$.
D_M^2	Squared Mahalanobis distance $\delta^T C + \delta$ from the external rate posterior mean.
T_9	Temperature in units of 10^9 kelvin .

Appendix C. Program and Data Manifest

Table C1. Principal programs

Program or path	Role in the investigation
analyze_bbn_lithium_linx_fr_network_2026-07-16.py	July LINX/FR network controls; reproduced the lithium problem and did not identify a D+He+Li pass.
analyze_su2_bbn_lithium_gate_2026-07-17.py	July SU2-style expansion/thermal gate for lithium; part of the negative control phase.
validate_bbn_lithium_linx_matrix_2026-07-17.py	Strict July validation matrix for the BBN lithium controls.
audit_bbn_lithium_linx_fr_scan_2026-07-17.py	Audit layer for the July LINX/FR scan.
run_cobaya_spd_cache_builder_2026_08_21.py	Built the validated SPD cache for the Planck PR4/NPIPE CamSpec likelihood.
postprocess_planck_pr4_full_nuisance.py	Post-processed the native full CMB nuisance chain and compared the registered lithium cell with the CMB posterior.
audit_planck_pr4_full_nuisance_qualification.py	Strict qualification audit: split chains, autocorrelation-aware ESS, registered-cell density, nuisance stability and half-chain stability.
start_planck_pr4_full_nuisance_background.ps1	Background launcher for the native Planck PR4 full nuisance MCMC.
start_planck_pr4_full_nuisance_ess_topup_background.ps1	Background launcher for the ESS top-up chain.
build_bbn_li_reports_20260911.py	This reproducible DOCX report generator.
run_bbn_lithium_nlo_thermal_2026_09_11.py	September matched NLO thermal gate for the registered mediator cell; checks pole-cut consistency, thermal self-energy proxies and NLO-size stability.
run_bbn_lithium_anchor_sensitivity_2026_09_11.py	September generic sensitivity grid over D/H, helium, stellar lithium and nuclear-rate proxy choices.
build_bbn_li_named_anchor_table.py	September named literature-anchor table and stress grid linking the registered candidate to specific observational anchors.
github_export/code/bbn_lithium/run_bbn_li_primat_rate_mc.py	PRIMAT small-network rate Monte Carlo importer and covariance bridge runner, including the N=5000 validated-response run.
github_export/code/bbn_lithium/diagnose_bbn_li_linx_covariance_eigenmode_validation.py	Direct LINX reruns along PRIMAT-derived covariance eigenmodes to validate local response linearity.
github_export/code/bbn_lithium/audit_bbn_li_covariance_stability_failure.py	N=1000 versus N=5000 covariance stability and abundance-failure decomposition.
github_export/code/bbn_lithium/diagnose_bbn_li_atlas_covariance_proxy.py	BBN sensitivity-atlas and NACRE-II covariance-proxy stress diagnostic.
github_export/code/bbn_lithium/diagnose_bbn_li_direct_cross_network.py	Direct abundance comparison between exact LINX points and matched PRIMAT reruns.
github_export/code/bbn_lithium/postprocess_bbn_li_direct_ddchain_worst_points.py	Worst-direct-point post-process for the fresh D-chain LINX/PRIMAT validation run.
diagnose_bbn_li_galah_hierarchical_depletion.py	Censored GALAH DR3 stellar-depletion/plateau model with a frozen GALAH DR4 validation.
rescore_bbn_li_archive_with_galah_posterior.py	Non-destructive archive-wide reclassification under the qualified GALAH lithium posterior.
diagnose_bbn_li_adaptive_shotgun.py	Adaptive 200,000-row mediator plus D-chain hypothesis search; not a confirmation sampler.
diagnose_bbn_li_combined_mediator_dchain.py	Direct LINX operator-split propagation of selected mediator plus D-chain candidates.
diagnose_bbn_li_candidate_external_covariance.py	External D-chain covariance audit and source-availability register.
update_bbn_li_problem_paper_2026_09_25_latest_results.py	Reproducible builder for this 25 September manuscript revision.

Table C2. Repository and output cross-reference

Object	Path or URL
Local output root	C:\Users\clive\Documents\Codex\2026-07-14\ok-contin\outputs\bbn_lithium_full_cmb_nuisance_20260821
Combined qualification audit	C:\Users\clive\Documents\Codex\2026-07-14\ok-contin\outputs\bbn_lithium_full_cmb_nuisance_20260821\qualification_audit_combined_ess_topup
Post-process folder	C:\Users\clive\Documents\Codex\2026-07-14\ok-contin\outputs\bbn_lithium_full_cmb_nuisance_20260821\postprocess
GitHub project	https://github.com/cliveboyd/Cosmology
GitHub BBN code	https://github.com/cliveboyd/Cosmology/tree/main/github_export/code/bbn_lithium
GitHub papers	https://github.com/cliveboyd/Cosmology/tree/main/github_export/docs/papers
September NLO thermal gate	C:\Users\clive\Documents\Codex\2026-07-14\ok-contin\outputs\bbn_lithium_nlo_thermal_gate_20260911
September generic anchor sensitivity	C:\Users\clive\Documents\Codex\2026-07-14\ok-contin\outputs\bbn_lithium_anchor_sensitivity_20260911
September named anchor sensitivity	C:\Users\clive\Documents\Codex\2026-07-14\ok-contin\outputs\bbn_lithium_named_anchor_sensitivity_20260911
N=5000 PRIMAT covariance bridge	plamb_runs/diagnostics/bbn_li_primat_rate_mc/bbn_li_primat_rate_mc_2026-09-12_primat_small_N5000_seed913_validated_response_report.md
N=5000 direct LINX eigenmode validation	plamb_runs/diagnostics/bbn_li_linx_eigenmode_validation/bbn_li_linx_eigenmode_validation_2026-09-12_small12_N5000_top8_pm1_pm2_report.md
Covariance stability and D/H failure audit	plamb_runs/diagnostics/bbn_li_covariance_stability_failure_audit/bbn_li_covariance_stability_failure_audit_2026-09-12_report.md
Atlas covariance-proxy stress test	plamb_runs/diagnostics/bbn_li_atlas_covariance_proxy/bbn_li_atlas_covariance_proxy_2026-09-12_atlas_response_proxy_N5000000_report.md
Fresh D-chain direct cross-network validation	plamb_runs/diagnostics/bbn_li_direct_cross_network/bbn_li_direct_cross_network_2026-09-12_fresh_ddhe3n_ddtp_overnight_report.md
Fresh D-chain worst-point post-process	plamb_runs/diagnostics/bbn_li_direct_cross_network/bbn_li_direct_cross_network_2026-09-12_fresh_ddhe3n_ddtp_overnight_worst_points_summary.md
GALAH hierarchical report	plamb_runs/diagnostics/bbn_li_galah_hierarchical_depletion/galah_hierarchical_depletion_validation_report.md
GALAH archive re-score	plamb_runs/diagnostics/bbn_li_galah_archive_rescore/bbn_li_galah_archive_rescore_report.md
GALAH 200k adaptive search	plamb_runs/diagnostics/bbn_li_adaptive_shotgun/bbn_li_adaptive_shotgun_2026-09-23_galah_posterior_extensive_200k_s1_report.md
Direct 60-row LINX batch	plamb_runs/diagnostics/bbn_li_combined_mediator_dchain/bbn_li_combined_mediator_dchain_2026-09-23_galah_posterior_extensive60_safebatch_s2_report.md
External covariance audit	plamb_runs/diagnostics/bbn_li_candidate_external_covariance/bbn_li_candidate_external_covariance_2026-09-24_galah60_external_covariance_report.md

Internal reproducibility references for the September 12 nuclear-rate update are listed in Appendix C and in the BBN research-hub register entries BBN-013 through BBN-022.

Appendix D1. External Reference Links Used in the Analysis

The following links are included so the publication draft connects each major claim boundary to an external source or numerical resource.

Reference	Link	Analytical use
Burns 2026 Physics Reports / BBN sensitivity atlas	Open source	The BBN-Li paper should mirror this logic: each external reference defines a gate or comparison, not merely a bibliography entry.
arXiv:2603.22414	Open source	Provides a direct model for linking reaction sensitivities and uncertainty budgets to the lithium problem.
BBN sensitivity atlas GitHub	Open source	Used as the main external response benchmark for the September 12 atlas import and NACRE-II stress branch.
Fields 2011 lithium review	Open source	Frames why a successful mechanism must reduce mass-7 without breaking the otherwise successful BBN light-element pattern.
Cybert Fields Olive Yeh 2016	Open source	Defines why broad expansion or baryon-density retuning is a high-risk route and why D/H and helium gates are non-negotiable.
Pitrou Coc Uzan Vangioni 2018 PRIMAT	Open source	Supports keeping PRIMAT cross-network comparison separate from a purely local LINX response result.
LINX code paper	Open source	Supports the local response-derivative/eigenmode-validation strategy, while still requiring direct mediator-aware propagation.
LINX arXiv preprint	Open source	Supports the technical decision to use LINX for fast response scans and direct finite-pull reruns.
ACROPOLIS electromagnetic-cascade framework	Open source	Defines why a late electromagnetic-transfer candidate needs cascade and threshold gates, not just a lithium-yield fit.
Planck 2018 cosmological parameters	Open source	Motivates replacing compressed CMB priors with the native PR4/NPIPE nuisance posterior used in September.
COBE FIRAS CMB spectrum	Open source	Defines the need for spectral-safety gates for any post-BBN electromagnetic energy release.
Khatri and Sunyaev 2012 CMB spectrum formation	Open source	Frames why mu-distortion and thermalisation checks are part of the candidate safety case.
CosmoVerse White Paper 2025	DOI	Added as an external gate reference: lithium improvement must survive D/H, Yp, He3/D, CMB, spectral, nuclear-rate and cross-dataset checks before promotion.
Wang et al. 2024 GALAH 3D-NLTE lithium catalogue	Open source	Large stellar-lithium catalogue used to build the empirical censored plateau/depletion model.
Launders, Giovanetti and Liu 2026 deuterium GP analysis	Open source	Highest-priority source for synchronised, correlated D-chain functional uncertainty.
Pisanti et al. 2021, Primordial Deuterium after LUNA	Open source	Published coefficient/error-budget basis used for the executable partial external covariance gate.
Mossa et al. 2020 LUNA deuterium-burning measurement	Open source	Experimental $d(p,\gamma)^3\text{He}$ anchor underlying the improved deuterium-rate programme.
Gomez Inesta, Iliadis and Coc 2017 Bayesian d+d rates	Open source	Bayesian treatment of the paired $d(d,n)^3\text{He}$ and $d(d,p)t$ reaction rates.
Moscoso et al. 2021 Bayesian $d(p,\gamma)^3\text{He}$ rate	Open source	Independent hierarchical Bayesian rate analysis for the third D-sensitive reaction.

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